

Lecture 12

Midterm Review

CS61B, Fall 2025 @ UC Berkeley

Josh Hug and Peyrin Kao

Question

“What situations (when defining classes and implementing interfaces) we have to include the generic input parameter (like in the signature, I noticed sometimes it’s both the class and the implemented interface, sometimes just 1).

What if the interface uses a differently named parameter for their generic definition (like E), we can still pass our parameter as T right?”

Question

“Related, but why is it – when creating the `ArraySetIterator` private class, which implements `Iterator` – that we use `implements Iterator<T>`, but don't write `ArraySetIterator<T>`? It's just `ArraySetIterator`, without the generic type (or any type!) in angle braces?”

“Re-explaining stack implementation from homework 5 “

“If it'll be useful to know, when and how to do/use the "helper" recursion”

- Example task: recursively sum a stack
- Example task 2: static void sumExcept34(int N), sums numbers from 1 to N except multiples of 3 and 4
- Example task 3: sorting an array sort(int[] x)

General principle: You can recurse directly when each subproblem is fully determined by the original argument(s) and you don't need to remember any extra context.

“explaining how to approach a doubly linked list problem with a sentinel node”



add first(1)



Question

“Would you please briefly recap each data type and structure's key words (LinkedList, DLList, IntList, Deque etc)?”

“A brief recap of the strengths and weaknesses (fast vs slow) of each data structure if possible!”

Question

“How should we break down Iterator exam problems—i.e. from which method should we start, how can we tell what to keep track of to maintain one's current position in the iteration? Would be great to have a step-by-step process. Thanks :)”

https://drive.google.com/file/d/1kTHkeOf-TnvlslIJU5GRF-GjdMhuq_2L/edit

https://drive.google.com/file/d/1-pPuHGuw0anqwnLh3nc_8z2Alqqcnpi5/view

“Comparator and Comparable and its usecases”